

POTPRESS · REQUIREMENTS

PotPress requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potpress		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (PPR-DDR-001); R10 target \$720, R6 band decided; status from PPR-CAL-001; proposed changes to R3 and R11
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

PotPress requirements

These requirements were checked by calculation at TRL 3 (PPR-CAL-001). They are not yet validated with a filter factory and will be revised after co-design sessions (see PPR-PRB-001). Three are **not met** on paper (R3 deflection, R10 cost and R11 for the QC rack), four are at risk (R2, R7, R8 and R9) and five are met on paper; see Table 2. Amish's decisions of 2026-09-25 are recorded in PPR-DDR-001: the R10 target is now \$720 and the R6 default band is decided. Changes to R3 and R11 are proposed, awaiting Amish.

The **reference filter** used throughout is the common flowerpot form: inner rim diameter 280 mm, inner depth 240 mm, wall 15 mm, flat rim about 345 mm across, about 12 L to the brim and about 10 L working volume (dimensions are estimates based on the 280 by 250 mm form reported by [Potters for Peace](#)).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Form the reference filter	Inner rim 280 mm, inner depth 240 mm, wall 15 mm, flat rim; geometry set by parameters in one source file	Model check; later measurement of pressed pots
R2	Even wall	Wall thickness within ± 1 mm around the circumference and from pot to pot; molds self-center to within 0.5 mm	Tolerance stack and guide design; later sectioned pots
R3	Pressing force and frame strength	20 t (196 kN) rated jack; frame, pins and platen carry 1.5 times the jack rating (294 kN) without yield and with deflection under 1 mm at the mold	Beam and pin calculation; later proof load
R4	Throughput	Press cycle of 6 min or less with two operators, so 50 or more pots per 6 h of pressing	Cycle breakdown; later timed trial
R5	Mold opening	Male mold lifts at least 30 mm clear of the female rim so the mold carriage slides out for loading and demolding; pressing stroke within one jack stroke	Kinematic layout
R6	Flow-rate QC	Rack tests 4 soaked filters at once; T-gauge reads flow over 1 h to 0.1 L/h; repeatability ± 0.1 L/h; acceptance band set per factory (default 1.0 to 2.5 L/h in the first hour, corrected to 25 °C; decided by Amish, 2026-09-25)	Gauge calibration by volume; later repeat tests
R7	Molds made locally	Molds cast from 3D-printed patterns printed on a 250 x 250 mm class printer; finishing with hand tools and a drill press; no lathe over 300 mm swing	Pattern split and finishing review with a foundry
R8	Frame built locally	Standard steel channel and plate; stick welding and drilling only; heaviest single part 40 kg or less for two-person handling	Part list and mass estimate
R9	Safe operation	Fixed guards on the sides and back, interlocked front gate; jack pump and release outside the guard; load pins must be in place before pressing	Hazard review; later guard check
R10	Affordable	Press plus QC rack \$720 or less in parts (raised from \$600 by Amish, 2026-09-25)	Priced BOM (bom/ bom.csv)
R11	Footprint	Press within 1.0 x 0.7 m floor area and 2.0 m height; QC rack within 0.8 x 0.5 m	Model check
R12	Product-safe materials	Faces that touch clay or test water are aluminum, food-grade polyethylene or stainless steel; no lead-based paint or oiled release agents on mold faces	Material list review

Table 2. Status against each requirement at TRL 3 (from PPR-CAL-001, not met first).

ID	TRL 3 value	Status
R3	No yield: beams 190 MPa, uprights 54 MPa, 60 mm pin 258 MPa (yield 650) at 294 kN. Deflection between the molds 1.84 mm at 294 kN, 1.23 mm at 196 kN, 0.61 mm at 10 t	Not met (deflection)
R10	Priced BOM \$926 (press \$845, QC rack \$81)	Not met , \$206 (29 %) over
R11	Press frame 840 x 640 mm, 970 mm deep with the fixed rail extension, 1,806 mm tall; QC rack 780 x 780 mm	Not met for the rack; press depth needs a hinged rail extension
R2	Wall ± 1.40 mm as cast; ± 0.43 mm with molds hand-finished to templates; coaxial ± 0.43 mm via the locating lip	At risk
R7	16 pattern segments for a 250 mm printer; cavity finishing without a lathe unproven	At risk
R8	Heaviest part as fabricated 38.9 kg (platen); welded frame 147 kg in one piece	At risk
R9	Guards, gate, jack-release interlock and pin-presence interlock in the BOM, not modeled	At risk
R1	Reference filter from <code>PARAMS</code> in <code>cad/src/model.py</code> : 12.30 L to the brim, 9.91 L working	Met on paper
R4	5.2 min cycle, 69 pots per 6 h	Met on paper, thin margin
R5	310 mm opening against 270 mm needed; 110 of 150 mm jack stroke	Met on paper
R6	4 stations; 0.1 L is 1.69 mm on the gauge; temperature correction to 25 °C	Met on paper; repeatability not verifiable at TRL 3
R12	Aluminum faces, polyethylene liners, HDPE buckets; scrap alloy must be lead-free	Met on paper

Proposed requirement changes (awaiting Amish)

- **R3**: judge deflection at the 10 t maximum working force (0.61 mm, met) and close the molds on a metal stop so frame stretch does not set the wall. Recommended (PPR-DDR-001 item 10).
- **R11**: allow 0.8 x 0.8 m for the four-station QC rack. Recommended (PPR-DDR-001 item 11).
- **R10**: raise the budget to about \$930, or cost the QC rack separately (PPR-DDR-001 item 14).

■ Assumptions

- Pressing force needed for a well-consolidated wall is not published in the sources found (one low-cost press forms round-bottom filters with a 2 t jack, [Henry, Maley and Mehta, 2013](#)); 5 to 10 t is assumed as the working range and the 20 t jack rating is used as the design load (estimate, to be confirmed with a partner factory).
- Mix follows the RDI-C recipe reported by [Rayner \(2009\)](#); the reference filter needs about 7.5 kg of mix per pot, below the 8 to 9.5 kg factories report (PPR-CAL-001).
- The default flow acceptance band of 1.0 to 2.5 L/h follows Nicaraguan practice and the Potters for Peace range of 1.5 to 2.5 L/h; each factory sets its own band.
- Allowable stress in structural steel is taken as 165 MPa (S275 with a factor of about 1.67).